Questions

Q1.

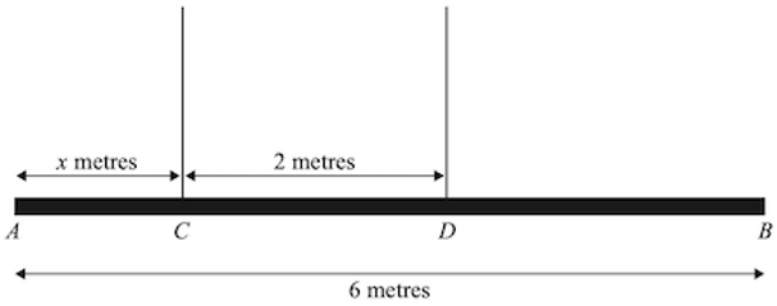


Figure 1

Figure 1 shows a beam AB with weight 24N and length 6 m.

The beam is suspended by two light vertical ropes. The ropes are attached to the points C and D on the beam where $AC = x$ metres and $CD = 2$ m.

The tension in the rope attached to the beam at C is double the tension in the rope attached to the beam at D.

The beam is modelled as a uniform rod, resting horizontally in equilibrium.

Find

(i) the tension in the rope attached to the beam at D.

(ii) the value of x.

(5)

(Total for question = 5 marks)

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(Q01 WME01/01, Oct 2023)

Q2.

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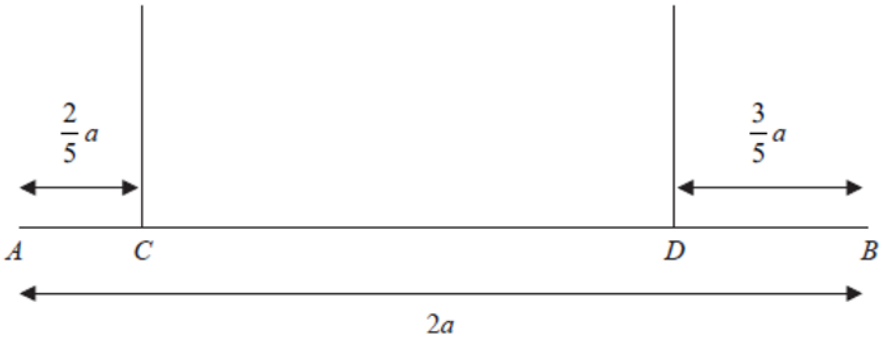


Figure 1

A uniform rod AB has length $2a$ and mass M. The rod is held in equilibrium in a horizontal position by two vertical light strings which are attached to the rod at C and D, where $AC = \frac{2}{5}a$ and $DB = \frac{3}{5}a$, as shown in Figure 1.

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A particle P is placed on the rod at B.

The rod remains horizontal and in equilibrium.

(a) Find, in terms of M, the largest possible mass of the particle P

(3)

Given that the mass of P is $\frac{3}{5}M$

(b) find, in terms of M and g, the tension in the string that is attached to the rod at C.

(3)

(Total for question = 6 marks)

(Q02 WME01/01, Oct 2022)

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Q3.

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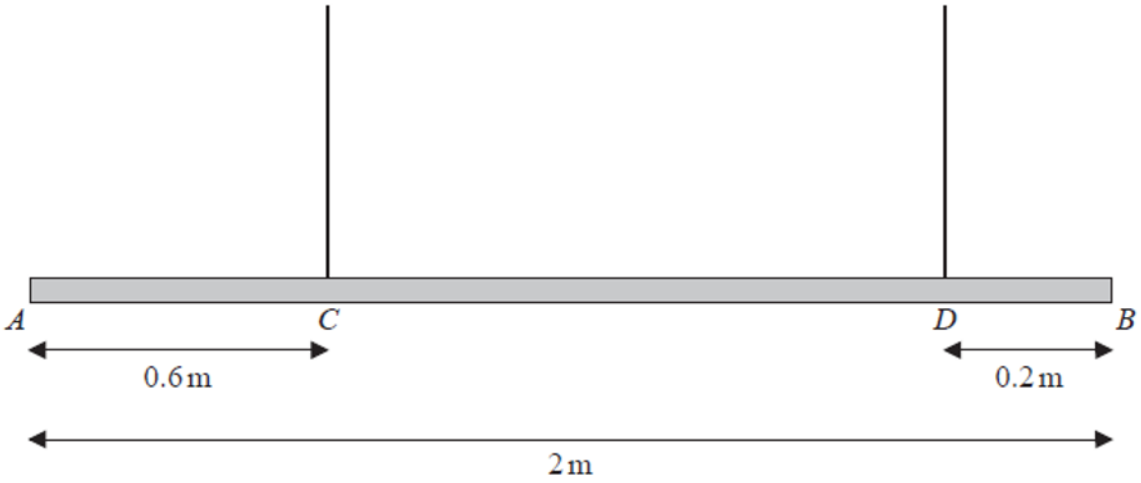


Figure 1

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Figure 1 shows a beam AB, of mass m kg and length 2 m, suspended by two light vertical ropes.

The ropes are attached to the points C and D on the beam, where $AC = 0.6$ m and $DB = 0.2$ m

The beam is in equilibrium in a horizontal position.

A particle of mass pm kg is attached to the beam at A and the beam remains in equilibrium in a horizontal position.

The beam is modelled as a uniform rod.

(a) Given that the tension in the rope attached at C is four times the tension in the rope attached at D, use the model to find the exact value of p.

(7)

The particle of mass pm kg at A is removed and replaced by a particle of mass qm kg at A.

The beam remains in equilibrium in a horizontal position but is now on the point of tilting.

(b) Using the model, find the exact value of q

(4)

(c) State how you have used the modelling assumption that the beam is uniform.

(1)

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(Total for question = 12 marks)

(Q04 WME01/01, June 2023)

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Q4.

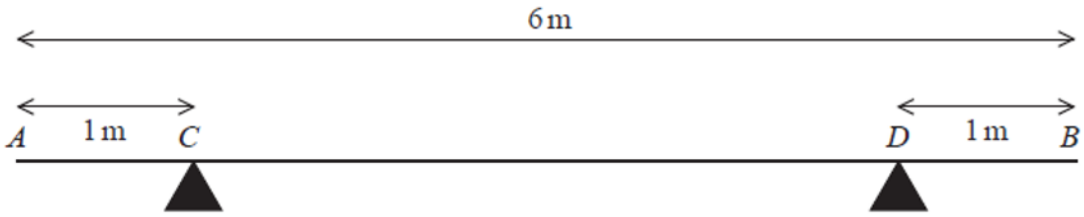


Figure 2

A metal girder AB has weight W newtons and length 6 m. The girder rests in a horizontal position on two supports C and D where $AC = DB = 1$ m, as shown in Figure 2.

When a force of magnitude 900 N is applied vertically upwards to the girder at A, the girder is about to tilt about D.

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When a force of magnitude 1500 N is applied vertically upwards to the girder at B, the girder is about to tilt about C.

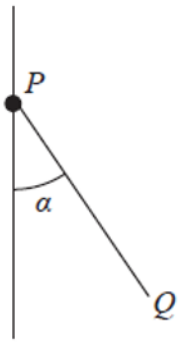
The girder is modelled as a non-uniform rod whose centre of mass is a distance x metres from A.

Find the value of x.

(Total for question = 6 marks)

(Q04 WME01/01, Jan 2021)

Q5.



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Figure 2

A small bead of mass 0.2 kg is attached to the end P of a light rod PQ. The bead is threaded onto a fixed vertical rough wire.

The bead is held in equilibrium with the rod PQ inclined to the wire at an angle α, where $\tan \alpha = \frac{4}{3}$

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, as shown in Figure 2.

The thrust in the rod is T newtons.

The bead is modelled as a particle.

(a) Find the magnitude and direction of the friction force acting on the bead when $T = 2.5$

(3)

The coefficient of friction between the bead and the wire is μ.

Given that the greatest possible value of T is 6.125

(b) find the value of μ.

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(7)

(Total for question = 10 marks)

(Q05 WME01/01, Oct 2021)

Q6.

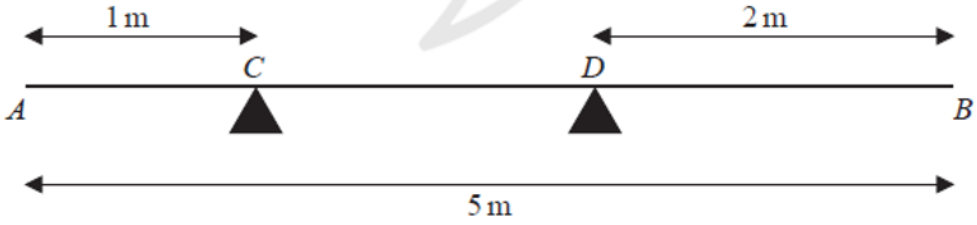


Figure 2

A uniform rod AB has length 5 m and mass 5 kg. The rod rests in equilibrium in a horizontal position on two supports C and D, where $AC = 1$ m and $DB = 2$ m, as shown in Figure 2.

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A particle of mass 10 kg is placed on the rod at A and a particle of mass M kg is placed on the rod at B. The rod remains horizontal and in equilibrium.

(a) Find, in terms of M, the magnitude of the reaction on the rod at C.

(3)

(b) Find, in terms of M, the magnitude of the reaction on the rod at D.

(3)

(c) Hence, or otherwise, find the range of possible values of M.

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(3)

(Total for question = 9 marks)

(Q05 WME01/01, June 2022)

Q7.

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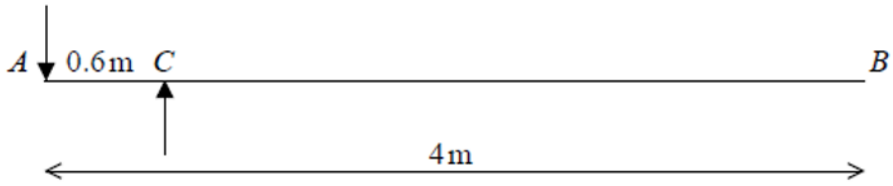


Figure 1

A diving board *AB* consists of a wooden plank of length 4m and mass 30kg. The plank is held at rest in a horizontal position by two supports at the points *A* and *C*, where *AC* = 0.6 m, as shown in Figure 1. The force on the plank at *A* acts vertically downwards and the force on the plank at *C* acts vertically upwards.

A diver of mass 50 kg is standing on the board at the end *B*. The diver is modelled as a particle and the plank is modelled as a uniform rod. The plank is in equilibrium.

(a) Find

(i) the magnitude of the force acting on the plank at *A*,

(ii) the magnitude of the force acting on the plank at *C*.

(6)

The support at *A* will break if subjected to a force whose magnitude is greater than 5000 N.

(b) Find, in kg, the greatest integer mass of a diver who can stand on the board at *B* without breaking the support at *A*.

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(3)

(c) Explain how you have used the fact that the diver is modelled as a particle.

(1)

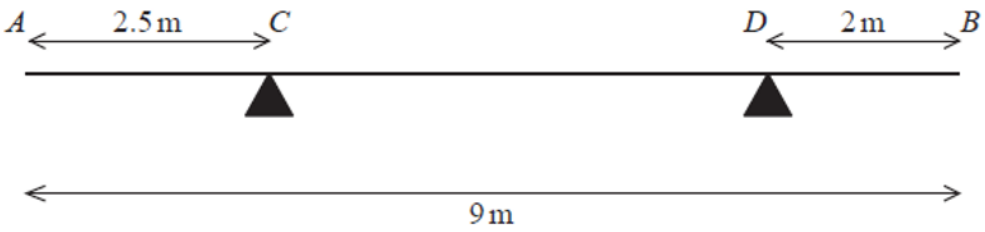
(Total for question = 10 marks)

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(Q04 WME11/01, Specimen papers)

Q8.



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Figure 1

A non-uniform rod *AB* has length 9 m and mass *M* kg.

The rod rests in equilibrium in a horizontal position on two supports, one at *C* where *AC* = 2.5 m and the other at *D* where *DB* = 2 m, as shown in Figure 1.

The magnitude of the force acting on the rod at *D* is twice the magnitude of the force acting on the rod at *C*.

The centre of mass of the rod is *d* metres from *A*.

Find the value of *d*.

(6)

(Total for question = 6 marks)

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Q9.

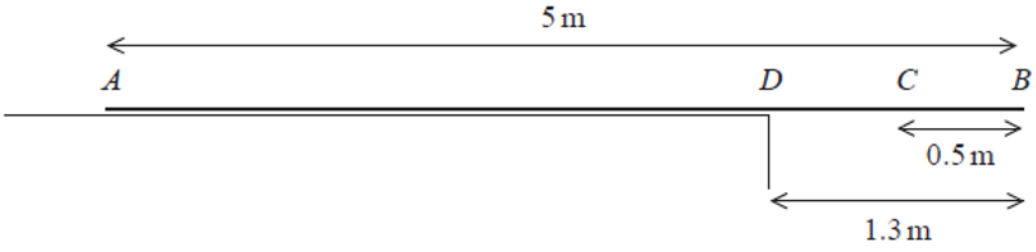


Figure 2

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A beam *ADCB* has length 5 m. The beam lies on a horizontal step with the end *A* on the step and the end*B* projecting over the edge of the step. The edge of the step is at the point *D* where *DB* = 1.3 m, as shown in Figure 2.

When a small boy of mass 30 kg stands on the beam at *C*, where *CB* = 0.5 m, the beam is on the point of tilting.

The boy is modelled as a particle and the beam is modelled as a uniform rod.

(a) Find the mass of the beam.

(3)

A block of mass *X* kg is now placed on the beam at *A*.

The block is modelled as a particle.

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(b) Find the smallest value of *X* that will enable the boy to stand on the beam at *B* without the beam tilting.

(3)

(c) State how you have used the modelling assumption that the block is a particle in your calculations.

(1)

(Total for question = 7 marks)

(Q03 WME01/01, Jan 2022)

Q10.

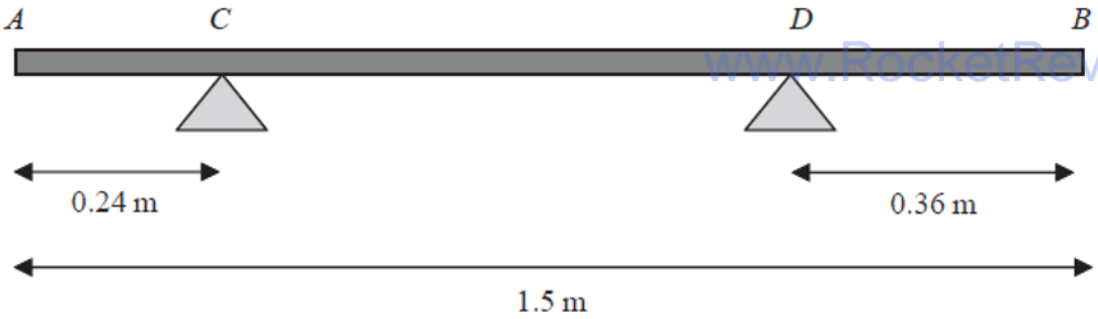


Figure 1

A branch *AB*, of length 1.5 m, rests horizontally in equilibrium on two supports.

The two supports are at the points *C* and *D*, where *AC* = 0.24m and *DB* = 0.36m, as shown in

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Figure 1.

When a force of 150 N is applied vertically upwards at B, the branch is on the point of tilting about C.

When a force of 225 N is applied vertically downwards at B, the branch is on the point of tilting about D.

The branch is modelled as a non-uniform rod AB of weight W newtons.

The distance from the point C to the centre of mass of the rod is x metres.

Use the model to find

(i) the value of W

(ii) the value of x

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(Total for question = 8 marks)

(Q04 WME01/01, Jan 2023)

Q11.

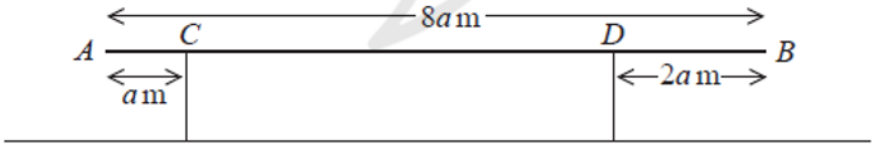


Figure 2

A non-uniform beam AB, of mass 60 kg and length $8a$ metres, rests in equilibrium in a horizontal position on two vertical supports. One support is at C, where $AC = a$ metres and the other support is at D, where $DB = 2a$ metres, as shown in Figure 2.

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The magnitude of the normal reaction between the beam and the support at D is three times the magnitude of the normal reaction between the beam and the support at C.

By modelling the beam as a non-uniform rod whose centre of mass is at a distance x metres from A,

(a) find an expression for x in terms of a.

(5)

A box of mass M kg is placed on the beam at E, where $AE = 2a$ metres.

The beam remains in equilibrium in a horizontal position.

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The magnitude of the normal reaction between the beam and the support at C is now equal to the magnitude of the normal reaction between the beam and the support at D.

By modelling the box as a particle,

(b) find the value of M.

(5)

(Total for question = 10 marks)

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(Q07 WME01/01, June 2021)

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